University Event Management System (UEMS)

Project Management Plan

Course: IT 612 Project Management in IT

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Executive Summary

This Project Management Plan (PMP) presents a comprehensive, professional-grade blueprint for the development of the University Event Management System (UEMS) at Qassim University. Qassim University, located in Buraidah in the Qassim region of Saudi Arabia, is one of the Kingdom's leading academic institutions under the College of Computer, Department of Information Technology. The university currently manages hundreds of academic and non-academic events including conferences, seminars, workshops, cultural festivals, and graduation ceremonies through predominantly manual and fragmented processes.

The existing manual approach to event management has created persistent operational challenges: scheduling conflicts between academic and non-academic events, poor communication among organizers and participants, budget mismanagement due to inadequate tracking mechanisms, and low efficiency in participant registration and attendance monitoring. These challenges have a direct impact on the university's administrative effectiveness and on the experience of its students, faculty, and external guests.

The UEMS is envisioned as a centralized, web-based digital platform that will automate the full lifecycle of event management from event creation, venue booking, and registration to participant communication, budget tracking, and post-event reporting. The system will serve multiple user roles including students, faculty, administrative staff, and university management.

This document covers all thirteen Project Management Knowledge Areas as defined by the Project Management Institute (PMI) PMBOK Guide, Seventh Edition, including: Integration Management, Scope Management, Schedule Management, Cost Management, Quality Management, Human Resource Management, Communications Management, Risk Management, Procurement Management, and Stakeholder Management.

The project is planned for completion within a 10-week academic timeline from February to April 2026, with a total estimated budget of SAR 285,000. The plan includes a comprehensive Work Breakdown Structure (WBS), Gantt chart, risk register with 10+ identified risks, RACI matrix, stakeholder register, and cost baseline.

This plan does not cover the actual software development of UEMS; rather, it provides the authoritative management framework that would govern the development project from initiation through closure, ensuring alignment with Qassim University's strategic objectives and the Saudi Vision 2030 initiative to digitize public service delivery.

1. Introduction

1.1 Organizational Background

Qassim University (QU) is a comprehensive public university established in 2004, headquartered in Buraidah, Qassim Region, Kingdom of Saudi Arabia. The university operates under the supervision of the Ministry of Education and serves a student population of over 60,000 across a wide range of undergraduate and postgraduate programs. The College of Computer within QU houses several departments including the Department of Information Technology (IT), which is responsible for developing technology-related competencies and digital solutions that serve both academic and administrative functions.

The Information Technology Department offers programs aligned with modern IT practices, preparing graduates to contribute to Saudi Arabia's growing digital economy in line with Vision 2030. The department regularly organizes academic events such as technical workshops, programming competitions, capstone project exhibitions, and research conferences.

1.2 Problem Statement

Currently, event management across Qassim University is conducted through a decentralized, manual process. Department administrators use a combination of phone calls, emails, paper-based forms, and shared spreadsheets to coordinate event logistics. This approach suffers from multiple systemic weaknesses:

- Scheduling conflicts arise when multiple events are booked in the same venue without a centralized booking system.
- Communication gaps between event organizers, participants, and university management lead to confusion and poor attendance.
- Budget mismanagement occurs because financial tracking is handled informally, making it difficult to reconcile expenditures against approved budgets.
- Participant tracking is inefficient; registration, attendance, and feedback collection rely on paper forms that are time-consuming to process.
- Event history and analytics are not systematically captured, preventing data driven decision making.

1.3 Project Purpose and Objectives

The purpose of this project is to plan the development of the University Event Management System (UEMS), a centralized digital platform that will resolve the identified challenges. The objectives of this project are:

1. Deliver a complete, PMBOK-aligned Project Management Plan for the UEMS development project within 10 weeks.
2. Define the project scope, schedule, budget, and quality standards with sufficient detail for execution.
3. Identify and plan for all major project risks, ensuring mitigation strategies are in place prior to development.

4. Establish stakeholder engagement and communication protocols to ensure alignment throughout the project.
5. Provide procurement and HR plans that support efficient resource acquisition and team management.

1.4 Scope of This Document

This Project Management Plan covers the planning phase of the UEMS project. It does not include actual system development, coding, or deployment. All sections follow the PMBOK Guide knowledge area framework and are designed to simulate a real-world IT project management scenario in an academic context.

2. Project Charter

2.1 Project Identification

Project Title	University Event Management System (UEMS)
Project Sponsor	Qassim University – Office of the Rector
Sponsoring College	College of Computer, Department of Information Technology
Project Manager	IT Project Management Team Lead
Start Date	February 1, 2026
Planned End Date	April 30, 2026
Duration	10 Weeks (2.5 Months)
Total Budget	SAR 285,000
Priority	High – Strategic Digital Transformation Initiative

2.2 Business Case

Qassim University manages hundreds of academic and social events each year across its expansive campus. The absence of a unified digital event management platform results in recurring inefficiencies estimated to cost the university over SAR 120,000 annually in lost productive time, duplicate resource allocation, and emergency budget adjustments. Additionally, the university's commitment to Saudi Vision 2030 and its institutional digital transformation strategy necessitates the modernization of core administrative processes, of which event management is a critical component.

The UEMS will directly improve operational efficiency, reduce administrative overhead, enhance student and faculty experience, and provide university management with real-time data for strategic decision-making. The return on investment is expected to be realized within 18 months of system deployment through measurable reductions in scheduling conflicts, manual processing time, and budget variances.

2.3 Problem Statement

Qassim University currently lacks a centralized system for managing academic and non-academic events. Manual coordination through emails, phone calls, and paper forms leads to scheduling conflicts, communication failures, budget mismanagement, and poor participant tracking, undermining the university's administrative effectiveness and stakeholder satisfaction.

2.4 SMART Objectives

SMART	Criterion	Objective
S	Specific	Develop a centralized web-based University Event Management System for QU.
M	Measurable	Reduce scheduling conflicts by 90% and manual processing time by 75% within 6 months of deployment.
A	Achievable	Complete planning within 10 weeks using available IT department expertise and university infrastructure.
R	Relevant	Aligns with QU digital transformation goals and Saudi Vision 2030 e-governance objectives.
T	Time-bound	Complete the full Project Management Plan by April 30, 2026.

2.5 High-Level Requirements

- Centralized event creation and management module accessible by authorized university staff.
- Online participant registration with automated confirmation and notification via email and SMS.
- Venue and resource booking with real-time conflict detection and resolution.
- Budget tracking module with approval workflows and variance reporting.
- Participant attendance management with QR-code-based check-in support.
- Post-event feedback collection and analytics dashboard.
- Role-based access control for students, faculty, event coordinators, and administrators.
- Integration with existing Qassim University student and HR information systems (MyQU portal).
- Mobile-responsive interface accessible via web browsers on desktop and mobile devices.
- Arabic and English bilingual interface to serve all university stakeholders.

2.6 Key Stakeholders

Stakeholder	Role	Interest
Rector's Office	Project Sponsor	Strategic oversight, institutional compliance, ROI
IT Department Head	Technical Sponsor	Technical feasibility, resource allocation
Project Manager	PM Lead	Project delivery within scope, time, cost
Development Team	Implementers	Clear requirements, realistic timelines

Event Coordinators	Primary Users	Ease of use, reliable scheduling tools
Students	End Users	Smooth registration, event discovery
Faculty Members	End Users	Efficient event organization support
QU IT Infrastructure	Technical Support	System integration, hosting, security

2.7 Assumptions and Constraints

Assumptions

- The university will provide access to the existing MyQU portal APIs for system integration.
- All required software licenses will be available within the budget.
- Faculty and staff will be available to participate in requirements-gathering sessions.
- The hosting infrastructure exists within the university IT environment.

Constraints

- The project planning phase must be completed within 10 academic weeks.
- The total project budget is capped at SAR 285,000.
- The system must comply with the Saudi National Cybersecurity Authority (NCA) data protection standards.
- All project documentation must be submitted in both Arabic and English.

2.8 Project Manager Authority

The Project Manager is authorized to: approve changes within a 10% variance of the approved budget (up to SAR 28,500); allocate team members to tasks; make decisions on technical approach within agreed architectural guidelines; escalate unresolved issues to the IT Department Head within 24 hours; and approve vendor selections for low-value procurement (under SAR 10,000) without sponsor approval.

3. Project Life Cycle Model

3.1 Selected Methodology

This project adopts a Waterfall (Predictive) life cycle model, which is appropriate given the clearly defined requirements, fixed academic deadline, and regulatory environment of a public university in Saudi Arabia. The waterfall model provides clear phase gates, structured approval workflows, and comprehensive documentation all aligned with the project's academic and institutional objectives.

The project will follow the five Process Groups defined by PMI PMBOK: Initiation, Planning, Execution, Monitoring & Controlling, and Closing.

3.2 Process Groups Overview

Phase	Duration	Key Activities
Initiation	Week 1	Charter approval, stakeholder identification, feasibility assessment
Planning	Weeks 2–5	Scope definition, WBS, schedule, cost, quality, risk, procurement, HR, communication plans
Execution	Weeks 6–8 (simulated)	System design, module development, integration, UAT
Monitoring & Control	Concurrent	Progress tracking, change control, risk monitoring, status reporting
Closing	Weeks 9–10	Final delivery, lessons learned, sign-off, documentation archiving

3.3 Phase Descriptions

Initiation Phase (Week 1)

The initiation phase formally authorizes the project. Activities include the development and approval of the Project Charter, identification of all key stakeholders, appointment of the Project Manager, and a preliminary feasibility review. Deliverables include the signed Project Charter and the initial Stakeholder Register. The phase gate exit criterion requires formal written approval from the Rector's Office and IT Department Head.

Planning Phase (Weeks 2–5)

The planning phase is the most comprehensive phase, producing all management plans. The project team conducts stakeholder requirements-gathering sessions, defines the detailed project scope, constructs the Work Breakdown Structure (WBS), develops the project schedule, estimates costs, and creates all subsidiary management plans. This phase's output is the complete, approved Project Management Plan which is the primary deliverable of this academic project.

Execution Phase (Weeks 6–8)

In a real deployment, the execution phase would involve the software development team implementing UEMS modules according to the approved design specifications. Work packages from the WBS would be assigned to team members, and progress would be tracked against the baseline. For this academic project, the execution phase is simulated through the development of realistic plans, prototypes, and decision artifacts.

Monitoring and Controlling (Continuous)

Throughout all phases, monitoring and controlling activities ensure the project remains on track. The Project Manager holds weekly status meetings, reviews the risk register, tracks schedule performance using Earned Value Management (EVM) metrics, and manages the change control process. Any deviation from baselines exceeding 10% triggers a corrective action report.

Closing Phase (Weeks 9–10)

The closing phase involves finalizing all project work, conducting a lessons-learned session, obtaining formal sign-off from the sponsor, archiving all project documentation, and delivering the final management report. The academic deliverable and the final presentation constitute the formal project closure artifacts.

3.4 Life Cycle Diagram

The following table illustrates the project life cycle phases, their timing, and primary outputs across the 10-week timeline:

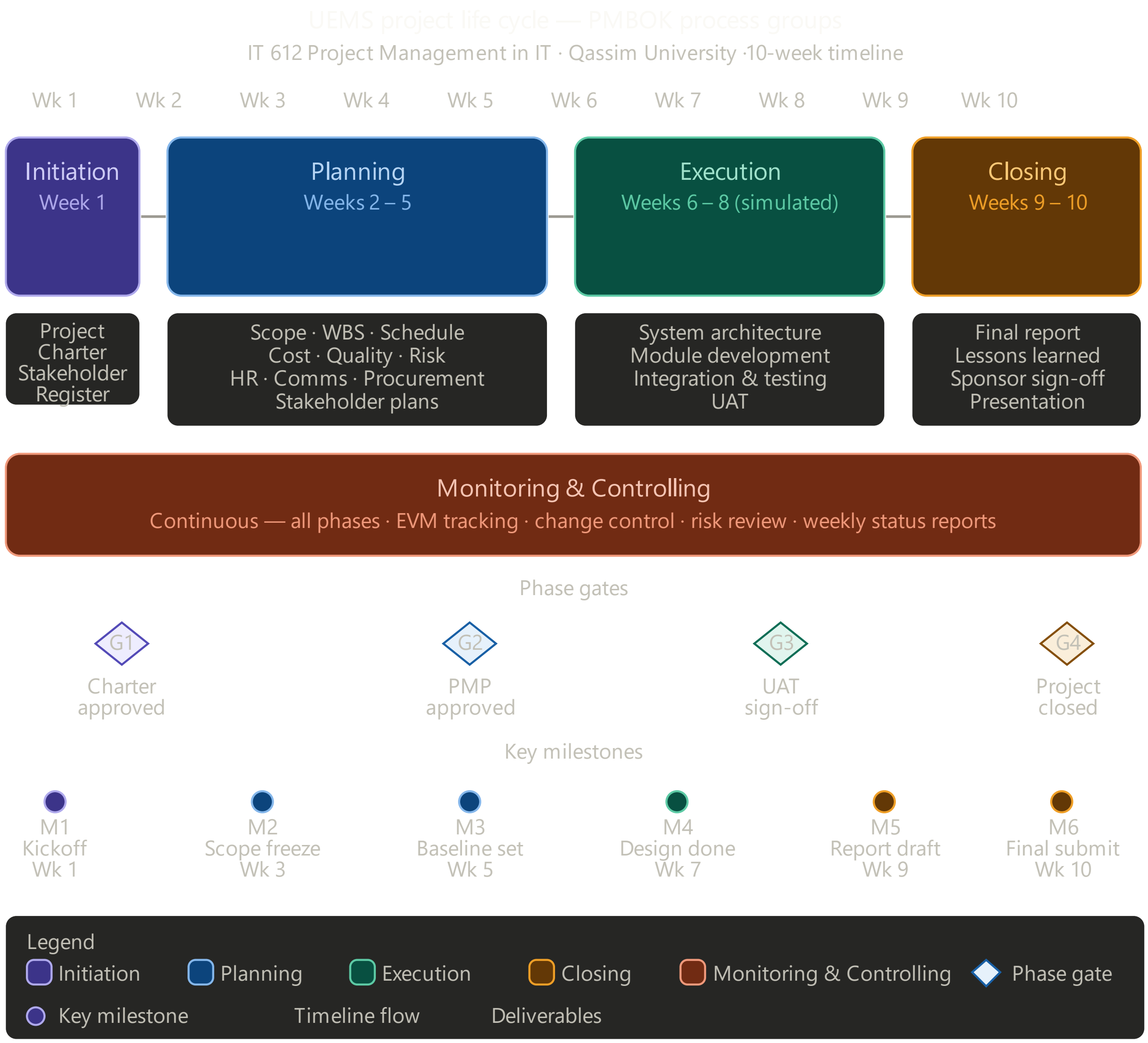

Figure 3.4.1: UEMS Life Cycle Diagram.

4. Project Integration Management

4.1 Overall Project Management Approach

Project Integration Management ensures that all project processes and activities are coordinated cohesively. For the UEMS project, integration management centers on the Project Manager as the single point of accountability for all management decisions. The Project Management Plan serves as the authoritative reference document for all project activities, and any changes must be processed through the formal change control process.

The Project Manager will conduct a weekly integration review meeting every Sunday from 10:00–11:00 AM, attended by all team leads. Meeting minutes will be circulated within 24 hours. The Project Manager will maintain a project dashboard updated weekly on the MyQU project portal, summarizing schedule, cost, risk, and quality status using a Red-Amber-Green (RAG) rating system.

4.2 Change Control Process

All changes to the project's approved baseline (scope, schedule, or cost) must follow the Integrated Change Control (ICC) process:

1. Change Requestor submits a Change Request Form (CRF) to the Project Manager, describing the change, rationale, and impact.
2. The Project Manager assesses the impact on scope, schedule, cost, quality, and risk within 48 hours.
3. For changes within the PM's authority threshold (≤10% budget variance), the PM approves or rejects and documents the decision.
4. For changes exceeding the PM's authority, the Change Control Board (CCB) comprising the PM, IT Department Head, and Finance representative convenes within 5 business days.
5. All approved changes are incorporated into the baseline documents and communicated to all stakeholders.
6. Rejected changes are documented with rationale and filed in the project records.

4.3 Configuration Management

All project documents and deliverables will be version controlled using a naming convention that includes the document type, version number, and date (e.g., UEMS_ScopeStatement_v1.2_20260215). Documents will be stored on the university's Microsoft SharePoint platform, organized by phase. Access permissions are defined in the Communications Management Plan. The Project Manager must approve any release of a document as a formal project deliverable.

4.4 Lessons Learned Process

Lessons learned will be collected throughout the project using a structured Lessons Learned Register. At the end of each phase, the team will hold a 30-minute retrospective to identify what went well, what could be improved, and what should be done differently. The final lessons learned report will be submitted to the IT Department's knowledge repository to benefit future projects at Qassim University.

5. Scope Management Plan

5.1 Scope Planning Approach

Scope management defines what is included in and excluded from the UEMS project. The scope will be defined through structured requirements-gathering sessions with key user groups, followed by formal documentation and stakeholder approval.

5.2 Detailed Scope Statement

In Scope

- Design and specification of the UEMS web application, including all functional modules.
- Event creation and management module.
- Participant registration module with email and SMS notifications.
- Venue and resource booking module with real-time conflict detection.
- Budget management module with multi-level approval workflow.
- Attendance tracking module with QR-code check-in capability.
- Feedback and survey module for post-event evaluation.
- Role-based user management (Admin, Coordinator, Faculty, Student roles).
- Integration specification with existing MyQU portal authentication (SSO).
- Bilingual interface specification (Arabic and English).
- System testing plan covering unit, integration, and user acceptance testing.

Out of Scope

- Actual software development and coding of the UEMS application.
- Hardware procurement and physical infrastructure setup.
- Training and deployment after system development.
- Integration with external third-party event platforms.
- Mobile application development (iOS/Android); the system is web-responsive only.

5.3 Work Breakdown Structure (WBS)

The WBS decomposes the total project scope into manageable work packages. The following represents the top three levels of the WBS:

L1	L2 Phase	L3 Deliverable	L4 Work Package	Est. Hours
1.0 Introduction	1.1 Initiation	1.1.1 Charter Development	Project Charter Document	16
	1.1 Initiation	1.1.2 Stakeholder ID	Stakeholder Register v1	8
	1.2 Planning	1.2.1 PMP Development	All subsidiary plans	120
	1.3 Monitoring	1.3.1 Status Reports	10× Weekly Reports	20
	1.4 Closing	1.4.1 Lessons Learned	Lessons Learned Report	8
2.0 Requirements	2.1 Elicitation	2.1.1 Stakeholder Interviews	Interview Notes (10 sessions)	20
	2.2 Documentation	2.2.1 Requirements Spec	SRS Document	40
	2.3 Validation	2.3.1 Requirements Sign-off	Signed approval	8
3.0 System Design	3.1 Architecture	3.1.1 Tech Stack Selection	Architecture Decision Record	16
	3.1 Architecture	3.1.2 System Architecture Doc	Architecture Diagram	24
	3.2 Database	3.2.1 Data Model Design	ERD + Schema	20
	3.3 UI/UX	3.3.1 Wireframes	30+ screen wireframes	40
	3.3 UI/UX	3.3.2 UI Prototype	Interactive Figma prototype	24
4.0 Development	4.1 Event Module	4.1.1 Event CRUD	Tested event management functions	80
	4.2 Registration	4.2.1 Registration Module	Registration + notification system	60
	4.3 Venue Booking	4.3.1 Booking Engine	Conflict detection module	50

	4.4 Budget Module	4.4.1 Budget Tracker	Budget management + approval	60
	4.5 Attendance	4.5.1 QR Check-in	QR generation + scanning module	40
	4.6 Feedback	4.6.1 Survey Module	Post-event survey tool	30
5.0 Integration	5.1 MyQU SSO	5.1.1 SSO Integration	Authentication integration	24
	5.2 Testing	5.2.1 Unit Testing	Unit test suite	40
	5.3 Testing	5.3.1 Integration Testing	Integration test results	30
	5.4 Testing	5.4.1 UAT Execution	UAT report + sign-off	50
6.0 Documentation	6.1 Technical Docs	6.1.1 API Documentation	API Reference Guide	16
	6.2 User Manuals	6.2.1 User Guide	Bilingual user manual	24
	6.3 Training	6.3.1 Training Materials	Training slide deck + videos	20
7.0 Deployment	7.1 Strategy	7.1.1 Deployment Plan	Go-live plan document	16
	7.2 Support	7.2.1 Support Plan	Post-launch support procedures	12

5.4 WBS Dictionary (Selected Work Packages)

ID	Work Package	Description	Responsible	Estimated Hours
1.2	PMP Development	All 13 knowledge area plans	PM + All Leads	120 hrs
2.2	Requirements Document	Functional & non-functional requirements specification	Business Analyst	40 hrs
3.1	Architecture Design	System architecture, tech stack, integration design	Lead Architect	60 hrs
4.1	Event Module	Development of event creation, listing, and management functions	Dev Team A	80 hrs
5.4	UAT	User Acceptance Testing with 20 test participants	QA Lead + Users	50 hrs

5.5 Scope Validation and Control

Scope validation will be conducted at the end of each major phase through a formal review meeting with the project sponsor and key users. Each deliverable will be assessed against its defined acceptance criteria before sign-off. Any work that falls outside the approved scope will be handled through the change control process described in Section 4.2. Scope creep will be actively monitored through weekly scope status reports and a formal scope baseline logged in the project management tool.

6. Schedule Management Plan

6.1 Schedule Development Approach

The project schedule is developed using a bottom-up estimation approach, decomposing the WBS work packages into individual activities, estimating durations, defining dependencies, and identifying the critical path. The schedule baseline is maintained in a Gantt chart format, and performance is tracked using Earned Value Management (EVM) metrics.

6.2 Activity List

ID	Activity	Start	End	Duration	Pred.
A1	Project Charter Development	Wk 1, Day 1	Wk 1, Day 5	5 days	—
A2	Stakeholder Identification	Wk 1, Day 3	Wk 1, Day 5	3 days	A1
A3	Requirements Gathering	Wk 2, Day 1	Wk 2, Day 5	5 days	A1, A2
A4	Scope Statement & WBS	Wk 3, Day 1	Wk 3, Day 5	5 days	A3
A5	Schedule Development	Wk 4, Day 1	Wk 4, Day 3	3 days	A4
A6	Cost Estimation & Budget	Wk 4, Day 3	Wk 5, Day 2	5 days	A4
A7	Risk Identification & Analysis	Wk 5, Day 3	Wk 6, Day 2	5 days	A4
A8	Quality Plan	Wk 5, Day 1	Wk 5, Day 3	3 days	A4
A9	HR & Communication Plans	Wk 6, Day 1	Wk 6, Day 3	3 days	A4
A10	Procurement Plan	Wk 6, Day 3	Wk 6, Day 5	3 days	A6
A11	System Architecture Design	Wk 6, Day 1	Wk 7, Day 5	8 days	A3
A12	Prototype/Wireframe Design	Wk 7, Day 1	Wk 8, Day 2	6 days	A11
A13	Plan Review & Revision	Wk 8, Day 1	Wk 9, Day 3	8 days	A1-A12
A14	Final Report Writing	Wk 9, Day 1	Wk 9, Day 5	5 days	A13
A15	Final Presentation Preparation	Wk 10, Day 1	Wk 10, Day 3	3 days	A14
A16	Submission & Presentation	Wk 10, Day 4	Wk 10, Day 5	2 days	A15

6.3 Activity Sequencing and Network Diagram

Activity sequencing uses Finish-to-Start (FS) dependencies as the primary relationship type. Key dependency chains are as follows:

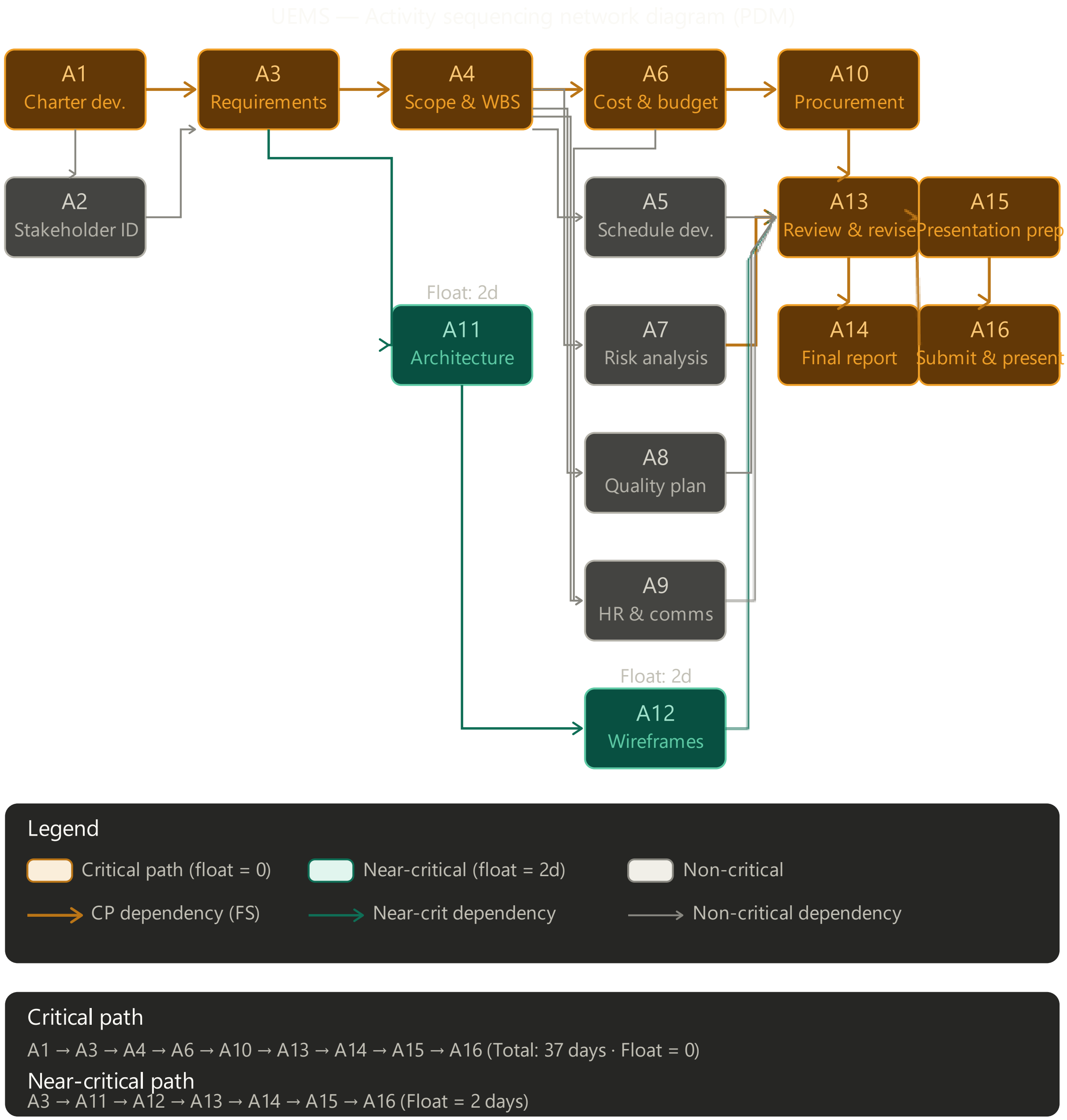

Figure 6.3.1: Activity Network Diagram using the Precedence Diagramming Method (PDM).

- Critical Path 1: A1 → A3 → A4 → A6 → A10 → A13 → A14 → A15 → A16 (Total: 37 days)
- Critical Path 2: A1 → A3 → A4 → A7 → A13 → A14 → A15 → A16 (Total: 35 days)
- Near-Critical: A11 → A12 runs in parallel with risk and quality planning (float: 2 days)

The critical path runs through requirements → scope → cost → procurement → revision → reporting → presentation, giving a total project float of zero on this chain. The Project Manager will monitor critical path activities weekly to prevent schedule slippage.

6.4 Gantt Chart Summary

Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
A1: Project Charter	■									
A2: Stakeholder ID	■									
A3: Requirements Gathering		■								
A4: Scope & WBS			■							
A5: Schedule Development				■						
A6: Cost & Budget				■	■					
A7: Risk Analysis					■	■				
A8: Quality Plan					■					
A9: HR & Communication						■				
A10: Procurement Plan						■				
A11: Architecture Design						■	■			
A12: Wireframe Design							■	■		
A13: Plan Review & Revision								■	■	
A14: Final Report Writing									■	
A15: Presentation Prep										■
A16: Submission										■

6.5 Critical Path Analysis

The Critical Path Method (CPM) analysis identifies

A1→A3→A4→A6→A10→A13→A14→A15→A16 as the critical path with zero total float. Any delay in these activities will directly delay the project completion. Activities on the near-critical path (A11, A12) have a total float of only 2 days and should be monitored closely. The project schedule includes a 3-day buffer in Week 10 to accommodate minor delays in report finalization.

6.6 Schedule Control

Schedule performance will be tracked using Earned Value Analysis (EVA). The key metrics used are: Schedule Performance Index (SPI = EV/PV) and Schedule Variance (SV = EV-PV). An SPI below 0.9 will trigger an escalation to the IT Department Head. Weekly schedule reports will be prepared and circulated every Sunday evening.

7. Cost Management Plan

7.1 Cost Estimation Approach

Cost estimation for the UEMS project uses a bottom-up estimation approach, building from individual work package cost estimates and aggregating to the total project budget. Estimates are based on Saudi market rates for IT professionals as of Q1 2026, historical data from similar QU IT projects, and vendor quotations for software licenses and hardware.

7.2 Detailed Cost Breakdown

Cat.	Item	Qty / Unit	Unit Cost (SAR)	Total (SAR)
Labor	Project Manager (10 weeks × 40 hrs/wk)	400 hrs	200 SAR/hr	**80,000**
Labor	Senior Developer × 2 (6 weeks × 40 hrs/wk)	480 hrs	175 SAR/hr	**84,000**
Labor	Business Analyst (4 weeks × 40 hrs/wk)	160 hrs	150 SAR/hr	**24,000**
Labor	UI/UX Designer (3 weeks × 40 hrs/wk)	120 hrs	140 SAR/hr	**16,800**
Labor	QA/Test Engineer (3 weeks × 40 hrs/wk)	120 hrs	130 SAR/hr	**15,600**
Software	Web Server License (Apache/Nginx — open source)	1 yr	0	**0**
Software	Database License (MySQL — community edition)	1 yr	0	**0**
Software	SMS Gateway API Subscription (12 months)	12 mo	1,500 SAR/mo	**18,000**
Software	Email Service API (SendGrid — business plan)	12 mo	750 SAR/mo	**9,000**
Hardware	Cloud Hosting / VPS Server (2-year commitment)	2 yrs	7,500 SAR/yr	**15,000**
Hardware	QR Code Scanner Hardware (check-in devices × 5)	5 units	800 SAR	**4,000**
Testing	UAT Coordination & Participant Compensation	Lump sum	—	**5,000**

Other	Contingency Reserve (10% of direct costs)	—	—	**27,140**
TOTAL				**298,540**

Note: The approved budget is SAR 285,000. The contingency reserve of SAR 27,140 will be drawn from the management reserve authorized by the project sponsor. Total project cost including all reserves is SAR 298,540.

7.3 Cost Baseline

The cost baseline represents the time-phased budget against which project expenditure will be measured. The cumulative S-curve expenditure distribution across the 10-week project is:

Item	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
Planned Spend (SAR '000)	8	18	22	28	26	42	48	36	24	16
Cumulative (SAR '000)	**8**	**26**	**48**	**76**	**102**	**144**	**192**	**228**	**252**	**268**

7.4 Cost Control

Cost performance will be measured using Earned Value Management (EVM). Key metrics monitored include: Cost Performance Index (CPI = EV/AC), Cost Variance (CV = EV - AC), and Estimate at Completion (EAC = BAC/CPI). A CPI below 0.9 will trigger a cost review meeting with the sponsor. Weekly cost reports will include actuals against the baseline with variance explanations.

8. Quality Management Plan

8.1 Quality Standards

The UEMS project will adhere to the following quality standards and frameworks:

- ISO/IEC 25010 (Software Quality Model) for system and software quality requirements.
- ISO 9001:2015 for overall project quality management processes.
- Saudi National Cybersecurity Authority (NCA) Essential Cybersecurity Controls (ECC-1:2018) for security quality.
- WCAG 2.1 Level AA for web accessibility, ensuring the system is usable by individuals with disabilities.
- QU Internal IT Project Quality Standards as defined by the Deanship of IT Affairs.

8.2 Quality Assurance (QA) Plan

Quality Assurance activities are proactive measures taken throughout the project to ensure processes are followed correctly. QA activities include:

- Weekly process audits to verify adherence to the Project Management Plan.
- Peer reviews of all major deliverables by at least one qualified reviewer before submission.
- Code review sessions for all developed modules, requiring sign-off from the Lead Architect before integration.
- Sprint-end retrospectives to evaluate process effectiveness and make improvements.
- Independent QA reviews at major phase gates before proceeding to the next phase.

8.3 Quality Control (QC) Plan

Quality Control activities are reactive inspection measures to verify that deliverable outputs meet defined standards. QC activities include:

- Functional testing of all UEMS modules against the requirements traceability matrix.
- Performance testing to verify the system supports at least 500 concurrent users without response degradation.
- Security testing including penetration testing and vulnerability scanning before UAT.
- Bilingual interface testing to verify correct rendering and functionality in both Arabic and English.
- UAT with a representative group of 20 users including students, faculty, and administrative staff.

8.4 Acceptance Criteria

Quality Dimension	Acceptance Criterion	Measurement Method
Functionality	100% of functional requirements implemented and verified	Requirements Traceability Matrix
Performance	Page load time < 3 seconds for 95% of requests under 500 concurrent users	JMeter Load Testing
Security	Zero critical vulnerabilities in penetration test report	OWASP ZAP Scan
Usability	System Usability Scale (SUS) score ≥ 75 (UAT)	UAT Survey
Reliability	System uptime ≥ 99.5% in staging environment over 2-week test	Server Monitoring Logs
Accessibility	WCAG 2.1 Level AA compliance	Automated + Manual Audit
Bilingual Support	100% of UI elements available in Arabic and English	Language Testing Checklist

8.5 Defect Management

Defects discovered during QC activities will be logged in the project defect tracking system, prioritized by severity (Critical, High, Medium, Low), assigned to the responsible developer, and resolved within defined SLAs: Critical defects within 24 hours, High within 3 business days, Medium within 1 week, Low before project closure.

9. Human Resource Management Plan

9.1 Team Structure

The UEMS project team is organized under a matrix structure, with team members reporting to both the Project Manager for project activities and to the IT Department Head for functional matters. The project team comprises:

Role	Name/Position	Availability	Key Responsibilities
Project Manager	Md Shakil Hossain IT Dept. Team Lead	100% (10 wks)	Overall planning, scheduling, stakeholder communication, reporting
Business Analyst	Dafaallh Alrayh Mustafa Senior IT Lecturer	75% (Wks 1–4)	Requirements gathering, scope documentation, WBS
Lead Architect	Ajmal Hamidi IT Infrastructure Specialist	60% (Wks 3–7)	System architecture, technology selection, design oversight
Senior Developer	Zayad Aldajan Final-year PhD Student	100% (Wks 5–9)	Event module, Registration module, Database design
UI/UX Designer	Biliaminu Ishola IT Department Graduate Assistant	80% (Wks 3–7)	Wireframes, interface prototypes, bilingual design
QA/Test Engineer	QA Specialist (contracted)	100% (Wks 7–9)	Test planning, execution, defect management, UAT coordination
IT Infrastructure Admin	QU IT Affairs Technician	25% (ongoing)	Server setup, network config, deployment environment

9.2 RACI Matrix

Activity	PM	BA	Arch	Dev A	Dev B	QA	Sponsor
Project Charter	R/A	C	—	—	—	—	I
Requirements Gathering	A	R	C	I	I	I	C
WBS Development	R/A	C	I	I	I	—	I

Schedule & Cost Planning	R/A	C	C	I	I	I	C
System Architecture Design	A	I	R	C	C	I	—
Module Development	A	I	C	R	R	I	—
Testing & QA	A	I	I	C	C	R	—
Stakeholder Reporting	R/A	I	I	I	I	I	I
Change Control Decisions	R/A	C	C	I	I	I	C
Project Closure	R/A	C	I	I	I	C	I

Legend: R = Responsible, A = Accountable, C = Consulted, I = Informed

9.3 Team Development Plan

To ensure effective team performance, the following development activities are planned:

- Project kickoff workshop (Week 1, 4 hours): team alignment, tool setup, communication norms.
- Requirement's facilitation training for the Business Analyst (Week 1, online).
- Agile estimation workshop for developers (Week 3, 2 hours).
- Information security awareness training for all team members (Week 2, mandatory, 1 hour online).
- Weekly 15-minute stand-up meetings to track progress and remove blockers.
- Mid-project team appreciation event (Week 6) to maintain morale and cohesion.

9.4 Conflict Resolution

Technical disagreements will be escalated to the Lead Architect for resolution. Resource conflicts will be escalated to the Project Manager, with final authority resting with the IT Department Head. All team members are expected to adhere to Qassim University's professional conduct standards and the Saudi Labor Law requirements.

10. Communication Management Plan

10.1 Communication Objectives

Effective communication is essential for project success. The Communication Management Plan ensures that all stakeholders receive accurate, timely, and relevant project information through appropriate channels and formats. All official communications will be in both Arabic and English as required by Qassim University's bilingual administrative policy.

10.2 Communication Matrix

Communication	Audience	Frequency	Method	Format	Owner
Project Status Report	Sponsor, IT Head	Weekly (Sunday)	Email + SharePoint	1-page PDF	Project Manager
Team Stand-up Meeting	Project Team	Weekly (Monday 9AM)	MS Teams Video Call	Meeting Minutes	PM / Rotating
Phase Gate Review	Sponsor + Stakeholders	End of each phase	In-person + Email	Presentation + Sign-off	Project Manager
Risk Log Update	PM + Leads	Bi-weekly	SharePoint	Risk Register	PM / Risk Owner
Change Request Notification	Relevant Stakeholders	As needed	Email	CRF + Impact Report	Project Manager
Issue Resolution Update	Affected Parties	As needed	Email / Teams	Issue Log Entry	Project Manager
Student/Faculty Bulletin	All QU Users	Monthly	University Portal	Bulletin Post	BA / Communications
Final Presentation	All Stakeholders	Week 10	In-person	Slides + Demo	Full Team

10.3 Communication Tools and Infrastructure

- Microsoft Teams: day-to-day team collaboration, file sharing, video meetings.
- SharePoint (QU IT Project Portal): official document repository, version control, phase deliverables.
- University Email (qu.edu.sa): formal communications with sponsor and external stakeholders.
- MyQU Portal: project announcements and stakeholder bulletins.
- WhatsApp Group (informal): quick team coordination and urgent notifications.

10.4 Reporting Structure

The reporting hierarchy flows upward from Team Members to Team Leads to the Project Manager to the IT Department Head to the Rector's Office. Reports are summarized at each level, with detailed data available on request. All reports follow the university's standard template format and are submitted in both Arabic and English.

10.5 Meeting Schedule

Meeting Type	Day/Time	Duration	Attendees	Output
Weekly Team Stand-up	Monday 9:00 AM	30 minutes	All team members	Action items, blockers
PM-Sponsor Sync	Wednesday 2:00 PM	45 minutes	PM, IT Head, Sponsor	Status update, decisions
Technical Review	Thursday 10:00 AM	60 minutes	PM, Arch, Developers	Technical decisions, design sign-off
Phase Gate Review	End of each phase	120 minutes	All stakeholders	Phase approval, next phase authorization
Risk Review	Every 2 weeks, Friday	30 minutes	PM, Risk Owners	Updated risk register

11. Risk Management Plan

11.1 Risk Management Approach

Risk management for the UEMS project follows a four-step iterative process: Identify → Analyze → Plan Responses → Monitor and Control. Risks are identified through structured brainstorming sessions with the project team, review of lessons learned from similar QU IT projects, and expert judgment from the IT Department. The Risk Register is the primary risk management artifact, maintained and updated bi-weekly.

11.2 Risk Probability-Impact Scales

Rating	Probability	Impact Description
1 – Very Low	< 10%	Negligible effect on schedule, cost, or quality
2 – Low	10–30%	Minor delay (< 1 day) or cost increase (< 2%)
3 – Medium	30–50%	Moderate delay (2–5 days) or cost increase (2–5%)
4 – High	50–70%	Significant delay (1–2 weeks) or cost increase (5–15%)
5 – Very High	> 70%	Critical impact; may compromise project success

11.3 Risk Register

ID	Risk Description	Prob.	Impact	Score	Rating	Category	Strategy	Owner
R01	Stakeholder requirements change after scope baseline is set	4	4	16	**High**	Scope	Mitigate: Enforce change control; freeze scope after Week 3	PM
R02	Key team member unavailability (illness or academic duty)	3	4	12	**High**	HR	Mitigate: Cross-train team; maintain skills matrix; identify backup resources	PM
R03	Integration failure with existing MyQU portal systems	4	5	20	**Critical**	Technical	Mitigate: Early API testing; involve QU IT from Week 2	Lead Arch

R04	Budget overrun due to unforeseen software licensing costs	2	4	8	**Medium**	Cost	Mitigate: Prefer open-source stack; budget contingency 10%	PM / Financ e
R05	Schedule delay from late requirements sign-off by stakeholders	4	4	16	**High**	Schedule	Mitigate: Set hard approval deadlines; escalate early	PM
R06	Data privacy breach in test environment	2	5	10	**High**	Security	Mitigate: Use anonymized test data; apply NCA ECC security controls	IT Admin
R07	Poor user adoption due to resistance to change	3	3	9	**Medium**	Change Mgmt	Mitigate: Involve users in UAT; conduct training sessions	BA
R08	Vendor delivery delay for QR scanner hardware	2	3	6	**Low**	Procurem ent	Mitigate: Order early (Week 3); identify alternative vendor	PM
R09	Technical debt from rushed development in later phases	3	4	12	**High**	Quality	Mitigate: Enforce code review standards; no shortcuts policy	Lead Arch
R10	Miscommunicati on between Arabic and English-speaking stakeholders	3	3	9	**Medium**	Communi cation	Mitigate: Bilingual meeting minutes; bilingual project documentation	BA
R11	Scope creep from informal feature requests	4	3	12	**High**	Scope	Mitigate: Formal change control; educate stakeholders on process	PM
R12	Inadequate testing time leading to defects at UAT	3	4	12	**High**	Quality	Mitigate: Parallel development/testin g; buffer in Weeks 8-9	QA Lead

11.4 Risk Monitoring

The Risk Register will be reviewed and updated at every bi-weekly risk review meeting. New risks identified during the project will be added immediately. Risk owners are responsible for implementing mitigation activities and reporting status at each review. The Project Manager will escalate any risk whose residual score exceeds 12 to the IT Department Head and project sponsor.

12. Procurement Management Plan

12.1 Procurement Strategy

The UEMS project will follow Qassim University's procurement policies, which are governed by the Saudi Government Tenders and Procurement Law (Royal Decree No. M/128). All procurement activities will be coordinated with the University's Finance and Procurement Department. The procurement strategy prioritizes open-source software solutions to minimize licensing costs, with commercial products selected only where open-source alternatives are insufficient.

12.2 Make-or-Buy Analysis

Component	Make	Buy	Rationale
Core UEMS Application	✓	—	Developed in-house using university IT staff and student team
Web Server (Apache/Nginx)	—	✓	Open-source; acquire at no cost
Database (MySQL Community)	—	✓	Open-source; no licensing cost
SMS Notification Service	—	✓	Specialized service; buy from established SMS gateway provider
Email Service (SendGrid)	—	✓	Reliable transactional email; buy cloud API subscription
QR Code Scanner Hardware	—	✓	Commodity hardware; purchase from IT supplier
UI/UX Design	✓	—	Internal designer with university branding capability
Security Penetration Testing	—	✓	Specialized service; engage qualified external cybersecurity firm
Training Materials Development	✓	—	Developed internally by BA and Documentation Lead

12.3 Vendor Selection Criteria

For items designated as Buy, vendors will be evaluated against the following weighted criteria:

Criterion	Weight	Evaluation Method
Technical capability and product quality	30%	Technical evaluation, product demos, proof-of-concept testing
Compliance with NCA security requirements	25%	Security documentation review, certifications
Price and value for money	20%	Competitive quotation
Vendor track record and references	15%	Reference checks, case studies from similar institutions
Support and SLA quality	10%	Review of support terms, response time commitments

12.4 Contract Types

The following contract types will be used for different procurement categories:

- SMS Gateway & Email Service: Fixed-Price contract with monthly subscription billing and 12-month term.
- Cloud Hosting: Fixed-Price contract with a 24-month commitment for reduced rates.
- QR Hardware: Fixed-Price Purchase Order, one-time.
- Penetration Testing: Time-and-Materials contract with a capped maximum fee of SAR 15,000.

12.5 Procurement Timeline

All procurement activities must be initiated by Week 4 at the latest, with vendor selection completed by Week 5 and all contracts signed by Week 6, allowing sufficient time for tool setup and integration before the development phase begins.

13. Stakeholder Management Plan

13.1 Stakeholder Identification Approach

Stakeholders were identified through structured workshops with the IT Department leadership, review of the university organizational chart, analysis of affected business processes, and consultation with the project sponsor. Stakeholders are categorized by their interest in the project outcome and their power to influence project decisions.

13.2 Stakeholder Register

Stakeholder	Role/Organization	Power	Interest	Current Engagement	Desired Engagement	Strategy
Rector's Office	Executive Sponsor	High	Medium	Aware	Supportive	Manage Closely
IT Department Head	Technical Sponsor	High	High	Supportive	Leading	Manage Closely
Project Manager	PM Lead	High	High	Leading	Leading	Monitor
Event Coordinators	Primary Users	Medium	High	Resistant	Supportive	Keep Satisfied
Students	End Users	Low	High	Neutral	Supportive	Keep Informed
Faculty Members	End Users / Organizers	Medium	Medium	Neutral	Supportive	Keep Informed
QU Finance Dept.	Budget Authority	High	Low	Aware	Supportive	Keep Satisfied
QU IT Infrastructure	Technical Support	Medium	High	Supportive	Leading	Manage Closely
NCA (Regulator)	Compliance Authority	High	Low	Unaware	Neutral	Keep Informed
External Vendors	Service Providers	Low	Low	Neutral	Neutral	Monitor

13.3 Power-Interest Grid

The power-interest grid categorizes stakeholders into four quadrants to guide engagement strategy:

HIGH POWER / HIGH INTEREST → Manage Closely	HIGH POWER / LOW INTEREST → Keep Satisfied
Rector's Office | IT Department Head | Project Manager | QU IT Infrastructure	QU Finance Department | National Cybersecurity Authority (NCA)
LOW POWER / HIGH INTEREST → Keep Informed	**LOW POWER / LOW INTEREST → Monitor**
Students | Faculty Members | Event Coordinators	External Vendors

13.4 Stakeholder Engagement Strategy

Manage Closely (High Power, High Interest)

These stakeholders will be engaged through weekly status meetings, direct involvement in phase gate reviews, early visibility into decisions, and proactive escalation of issues. The Project Manager will maintain a dedicated relationship log for each of these stakeholders.

Keep Satisfied (High Power, Low Interest)

QU Finance and the NCA will receive monthly summary reports and formal compliance documentation. They will be engaged at project milestones and when decisions requiring financial or regulatory approval are needed.

Keep Informed (Low Power, High Interest)

Students, faculty, and event coordinators will be kept informed through monthly bulletins on the MyQU portal, participation in UAT sessions, and regular feedback surveys. Their input will be actively solicited during requirements gathering and UAT.

Monitor (Low Power, Low Interest)

External vendors will be monitored through procurement processes and contract performance management. They will receive project updates only as relevant to their contracted deliverables.

13.5 Resistance Management

Event coordinators have been identified as potentially resistant to system change, as the UEMS will significantly alter their current workflows. The resistance management strategy includes: involving 3 senior coordinators as 'UEMS Champions' from the planning phase, conducting hands-on demonstrations of system prototypes, emphasizing efficiency benefits, and providing dedicated training resources before system deployment.

14. Conclusion

This Project Management Plan represents a comprehensive, PMBOK-aligned framework for the planning, governance, and delivery of the University Event Management System (UEMS) at Qassim University, College of Computer, Department of Information Technology.

The plan addresses all thirteen Project Management knowledge areas, providing a detailed and actionable blueprint that covers: the project charter and business case, the full project life cycle from initiation through closure, scope boundaries and a three-level Work Breakdown Structure, a 10-week schedule with critical path identification, a detailed cost breakdown and budget baseline of SAR 285,000, comprehensive quality standards and acceptance criteria aligned with ISO 25010 and NCA security requirements, a team structure with RACI matrix and development plan, a multi-channel communication strategy in both Arabic and English, a 12-risk risk register with probability-impact analysis and mitigation strategies, a procurement plan aligned with Saudi government procurement law, and a stakeholder engagement strategy using the power-interest grid framework.

The UEMS project directly supports Qassim University's strategic objectives and the broader Saudi Vision 2030 initiative to digitize public services and enhance institutional efficiency. By replacing fragmented manual processes with a centralized, bilingual, role-based digital platform, the university will eliminate scheduling conflicts, improve participant experience, and gain data-driven insights into its event ecosystem.

The success of this project depends on disciplined execution of this management plan, active stakeholder engagement, rigorous risk monitoring, and the commitment of the entire project team to delivering a high-quality outcome within the defined constraints of scope, schedule, and budget. With the robust governance framework established in this document, the UEMS project is well-positioned to deliver lasting value to Qassim University and all its stakeholders.